Solution: A set of geometric random variables can be used here. So expected rolls $= k \ln(k)$ (Done in class).

(c) Let $k = 3$ and the die is biased, i.e., $\mathbf{P}(1) = \mathbf{P}(3) = \frac{1}{4}$ and $\mathbf{P}(2) = \frac{1}{2}$. Over expectation how many times you need to roll the die until you see every number from 1 to 3 at least once on its upward face. **(5)**

Solution: Lets define three Bernoulli/indicator random variables as X_i for every $i \in [3]$ to count the number of rolls needed to witness an unseen face value i. So $X_1 = X_3 = 1$ with probability $1/4$ and $X_2 = 1$ with probability $1/2$. Let $X = \sum_{i=1}^{3} X_i$, then $\mathbf{E}[X] = 1$. We witness an unseen face i in the first roll (trivial).

Consider three geometric random variables Y_i to analyze the number of rolls to witness the second unseen face value. The random variable Y_1 counts the number of rolls until we witness another unseen face value, while the first unique face was 1. The $P(\text{face value 1}) = 1/4$, so we have $P(\text{face value 2 or 3}) = 1/2 + 1/4 = 3/4$. Now, $P(\text{second unseen face value is 2 or 3}) = P(\text{first seen face value was 1}) = 1/4$. So, $Y_1 = j$ (i.e., witness second unseen face value 2 or 3 in j rolls) with probability $P(\text{first seen face value was 1}) \cdot P(\text{face value 1})^{j-1} \cdot P(\text{face value 2 or 3})$. Formally,

$$Y_1 = \begin{cases} j & \text{with probability } \frac{1}{4}\frac{1}{4}^{j-1}\frac{3}{4} \end{cases}$$

Similary we define the other two random variables as follows,

$$Y_2 = \begin{cases} j & \text{with probability } \frac{1}{2}\frac{1}{2}^{j-1}\frac{1}{2} \text{ and,} \end{cases}$$

$$Y_3 = \begin{cases} j & \text{with probability } \frac{1}{4}\frac{1}{4}^{j-1}\frac{3}{4}. \end{cases}$$

Let $Y = \sum_{i=1}^{3} Y_i$. Then $\mathbf{E}[Y] = \mathbf{E}[Y_1] + \mathbf{E}[Y_2] + \mathbf{E}[Y_3] = \frac{1}{4}\frac{4}{3} + \frac{1}{2}\frac{2}{1} + \frac{1}{4}\frac{4}{3} = \frac{5}{3}$.

Consider three more geometric random variables Z_i to analyze the number of rolls until we witness the final unseen face value $i \in [3]$. Here Z_1 is the random variable that counts the number of roll until we see the final unseen face value 1. To analyze the probability distribution of Z_1, we have, $P(\text{final unseen face value is 1}) =$
$P(\text{first seen face value was 2}) \cdot P(\text{second seen face value was 3}|\text{face value 2 or 3}) +$
$P(\text{first seen face value was 3}) \cdot P(\text{second seen face value was 2}|\text{face value 3 or 2})$
$= P(\text{first seen face value was 2}) \cdot \frac{P(\text{second seen face value was 3})}{P(\text{face value 1 or 3})} +$
$P(\text{first seen face value was 3}) \cdot \frac{P(\text{second seen face value was 2})}{P(\text{face value 3 or 2})} = \frac{1}{2}\frac{\frac{1}{4}}{\frac{1}{2}} + \frac{1}{4}\frac{\frac{1}{2}}{\frac{3}{4}} = \frac{5}{12}$. So,

$$Z_1 = \begin{cases} j & \text{with probability } \frac{5}{12}\frac{3}{4}^{j-1}\frac{1}{4}. \end{cases}$$

Now, due to symmetry for Z_3 we have, $P(\text{final unseen face value is 3}) = \frac{5}{12}$ and for Z_2 we have, $P(\text{final unseen face value is 2}) =$
$P(\text{first seen face value was 1}) \cdot \frac{P(\text{second seen face value was 3})}{P(\text{face value 2 or 3})} +$
$P(\text{first seen face value was 3}) \cdot \frac{P(\text{second seen face value was 1})}{P(\text{face value 1 or 2})} = \frac{1}{4}\frac{\frac{1}{4}}{\frac{3}{4}} + \frac{1}{4}\frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{6}$. So,

$$Z_2 = \begin{cases} j & \text{with probability } \frac{1}{6}\frac{1}{2}^{j-1}\frac{1}{2}, \text{ and,} \end{cases}$$

$$Z_3 = \begin{cases} j & \text{with probability } \frac{5}{12}\frac{3}{4}^{j-1}\frac{1}{4}. \end{cases}$$

Let $Z = \sum_{i=1}^{3} Z_i$, then $\mathbf{E}[Z] = \mathbf{E}[Z_1] + \mathbf{E}[Z_2] + \mathbf{E}[Z_3] = \frac{5}{12}\frac{4}{1} + \frac{1}{6}\frac{2}{1} + \frac{5}{12}\frac{4}{1} = \frac{11}{3}$. Finally, the expected number of rolls to witness all the face values at least once is, $\mathbf{E}[X + Y + Z] = 1 + \frac{5}{3} + \frac{11}{3} = \frac{19}{3}$.